

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Lab 2: Spatial Occurrence Data, Mapping, and BAM Reasoning

Today you will turn occurrence records into spatial objects, make a reproducible map, and ask what that map can and cannot support as evidence about a species' geographic distribution.

What you need

- Your revised Lab 1 project and a private GitHub repository.
- R and RStudio. Install `sf`, `ggplot2`, `rnatrualearth`, and `rnatrualearthdata` if needed.
- Your Lab 1 script must create `occ_retained` before you begin this lab.

Part A - Setup

1. **Open your Lab 1 R Project.** Do not make a second project or copy your raw data elsewhere.
2. **Create a new script.** Save it as `analysis/02_spatial_occurrence_map.R`.
3. **Install packages once, then load them in every new R session.**

```
# Run once, then comment out
install.packages(c("sf", "ggplot2", "rnatrualearth", "rnatrualearthdata"))

# Run at the start of this script
library(sf)
library(ggplot2)
library(rnatrualearth)
```

4. **Bring forward your Lab 1 decisions.** Run your Lab 1 script from the project root, then confirm that both the raw and retained occurrence objects exist. The code below gives the process for checking the retained occurrences.

```
source("analysis/01_occurrence_audit.R")
stopifnot(exists("occ_retained"))
nrow(occ_retained)
```

If this step fails, return to Lab 1. Do not create a new filtered dataset by hand or overwrite the raw CSV.

Part A continued - Represent records as spatial points

An sf object stores an attribute table and a geometry column together. The coordinate reference system (CRS) tells R how the coordinates relate to locations on Earth. Here, the occurrence records are stored as longitude and latitude using WGS 84 (EPSG:4326). WGS 84 is a geographic coordinate reference system: locations are described using angular coordinates in degrees of longitude and latitude. Importantly, assigning `crs = 4326` tells R what the existing coordinates mean. It does not change the coordinates. Changing an object from one CRS to another requires a transformation using `st_transform()`.

```
# Convert the original records with coordinates to an sf object
occ_raw_sf <- st_as_sf(
  occ_raw,
  coords = c("decimalLongitude", "decimalLatitude"),
  crs = 4326,
  remove = FALSE
)

# Convert the Lab 1 retained records to an sf object
occ_sf <- st_as_sf(
  occ_retained,
  coords = c("decimalLongitude", "decimalLatitude"),
  crs = 4326,
  remove = FALSE
)
```

Because `remove = FALSE`, the original longitude and latitude columns remain in the attribute table. The geometry column also stores the spatial coordinates.

```
# View the first several coordinates stored in the geometry
head(st_coordinates(occ_sf))
```

These values are longitude and latitude in degrees because the object uses EPSG:4326.

Inspect the spatial object

Before analyzing spatial data, check what kind of spatial object you actually created.

```
# Coordinate reference system
st_crs(occ_sf)

# Geometry type
st_geometry_type(occ_sf)

# Geographic extent of the data
st_bbox(occ_sf)
```

You should see a WGS 84 CRS, POINT geometry, and a bounding box giving minimum and maximum x and y coordinates. These are checks on the meaning and structure of your data, not merely technical output.

Quick spatial plotting

sf objects have a built-in plotting method, but `plot()` is useful when you want to quickly inspect a spatial object.

```
plot(occ_sf["year"])
```

Because year is included, R uses that attribute to symbolize the points. Now plot only the geometry:

```
plot(st_geometry(occ_sf))
```

Think about: What information disappeared when you used `st_geometry()`?

Build maps with ggplot2

For more reproducible and customizable figures, we will use `ggplot2`. A `ggplot` figure is built in layers.

```
# Create a plotting canvas and add the occurrence points
ggplot() +
  geom_sf(data = occ_sf)
```

`ggplot()` creates the plotting canvas. `geom_sf()` adds a spatial layer. Because `occ_sf` already contains geometry, we do not need to specify `x = longitude` and `y = latitude`; the geometry column tells `ggplot2` where the features belong.

Set the appearance of a layer

```
ggplot() +
  geom_sf(
    data = occ_sf,
    color = "black",
    size = 1.5
  )
```

Here, `color` and `size` are outside `aes()`, so every occurrence receives the same appearance.

Map an appearance to a variable

```
ggplot() +  
  geom_sf(  
    data = occ_sf,  
    aes(color = year),  
    size = 1.5  
  )
```

Here, year is inside aes(), so color is connected to information in the data. ggplot2 also creates a legend automatically.

Add geographic context

The occurrence points have locations, but we currently have little geographic context. Add country polygons from rnatualearth.

```
world <- ne_countries(  
  scale = "medium",  
  returnclass = "sf"  
)  
  
st_crs(world)
```

world is also an sf object, but its geometry consists of polygons rather than points.

```
ggplot() +  
  geom_sf(  
    data = world, fill = "grey95",  
    color = "grey70", linewidth = 0.2) +  
  geom_sf(  
    data = occ_sf, color = "red",  
    size = 1.2, alpha = 0.7) +  
  labs(title = "C. triseriatus occurrence records") +  
  theme_minimal(base_size = 12)
```

Layers are drawn in the order they appear in the code. The country polygons are drawn first and the occurrence points are drawn on top.

Compare original and retained records

Layer the original beneath the retained records.

```
ggplot() +  
  geom_sf(data = world,  
    fill = "grey95", color = "grey70", linewidth = 0.2) +  
  geom_sf(data = occ_raw_sf, size = 1, alpha = 0.7, color = "red") +  
  geom_sf(data = occ_sf, size = 1, alpha = 0.7, color = "blue3") +  
  labs(title = "C. triseriatus occurrence records") +  
  theme_minimal(base_size = 12)
```

Control the map extent

Save the ggplot as an object and create a bounding box. The bounding box tells about the geographic extent of the original records.

```
# save your map to an object

map_output <- ggplot() +
  geom_sf(data = world, fill = "grey95",
    color = "grey70", linewidth = 0.2) +
  geom_sf(data = occ_raw_sf,
    color = "red", size = 1, alpha = 0.7) +
  geom_sf(data = occ_sf,
    color = "blue3", size = 1, alpha = 0.7) +
  labs(title = "C. triseriatus occurrence records") +
  theme_minimal(base_size = 12)

# use the bounding box of the raw occurrence records
bb <- st_bbox(occ_raw_sf)

bb
```

Use that information to focus the map around the occurrence records.

```
# create padding
x_pad <- max(0.25, 0.10 * (bb["xmax"] - bb["xmin"]))
y_pad <- max(0.25, 0.10 * (bb["ymax"] - bb["ymin"]))

# apply the new extent with padding to the map
map_output +
  coord_sf(
    xlim = c(bb["xmin"] - x_pad, bb["xmax"] + x_pad),
    ylim = c(bb["ymin"] - y_pad, bb["ymax"] + y_pad),
    expand = FALSE)
```

`coord_sf()` changes the viewing window of the figure. It does not remove records, filter the dataset, or change the coordinates.

Save the map

A ggplot is itself an R object. We can create it, modify it, inspect it, and then save it reproducibly.

```
ggsave("output/occurrence_map_sf.png", plot = map_output, width = 7, height = 6, dpi = 300)
```

Open the PNG in your output/ folder and confirm that the script created the expected figure.

Geographic versus projected coordinates

So far, the occurrence data use WGS 84 (EPSG:4326). Their coordinates are longitude and latitude measured in degrees:

```
st_crs(occ_sf)
st_coordinates(occ_sf)[1:5, ]
```

A projected coordinate reference system transforms locations from the curved surface of Earth onto a flat coordinate system. Projected coordinates are usually expressed in linear units such as meters, which makes them more useful for many operations involving distance, area, and spatial neighborhoods. Different projections preserve different geographic properties, so there is no single projection that is appropriate for every analysis. For demonstration, transform the occurrence records to WGS 84 / UTM Zone 14N (EPSG:32614):

```
occ_projected <- st_transform(occ_sf, 32614)
st_crs(occ_projected)
st_coordinates(occ_projected)[1:5, ]
st_bbox(occ_projected)
```

Compare the projected coordinates with the original geographic coordinates:

```
st_coordinates(occ_sf)[1:5, ]
```

The numbers changed because the coordinate systems use different ways of describing location: EPSG:4326 uses longitude and latitude in degrees; EPSG:32614 uses x and y coordinates in meters. The occurrence locations themselves did not move on Earth.

Important distinction: `crs = 4326` describes what existing coordinates mean; `st_transform()` calculates new coordinates in another CRS.

Buffering for M accessible area

In the BAM framework, **M** represents the geographic area that has been accessible to the species through movement or dispersal over the relevant period of time. We can use a spatial operation to represent a simple **hypothesis** about accessibility. Because our projected coordinates are measured in meters, we can create a distance buffer around the occurrence records.

```
# create a 100-km buffer around each retained occurrence point
# projected coordinates are in meters, so 100 km = 100,000 m
m_buffer <- st_buffer(occ_projected, dist = 100000)

# combine overlapping buffers into one accessible area
m_area <- st_union(m_buffer)
```

```
# plot the hypothesis about M

ggplot() +
  geom_sf(data = world, fill = "grey95", color = "grey70") +
  geom_sf(data = st_transform(m_area, 4326), fill = "lightblue",
    color = "blue", alpha = 0.3) + geom_sf(data = occ_sf, color = "black") +
  coord_sf( xlim = c(bb["xmin"] - x_pad, bb["xmax"] + x_pad),
    ylim = c(bb["ymin"] - y_pad, bb["ymax"] + y_pad), expand = FALSE) +
  theme_minimal(base_size = 12)
```

Of course, keep in mind, this is not an actual biologically defensible estimate of M. It is a spatial representation of **one assumption** about accessibility. A biologically defensible M would require information about dispersal ability, barriers, history, the time period of interest, and the natural history of the species.

Think about: What assumptions are we making when we draw the same 100-km buffer around every occurrence record?

Part B - Do it yourself

Complete the following in `analysis/02_spatial_occurrence_map.R` and in your README. Use R help and the troubleshooting notes when you get stuck.

Problem 1: Inspect the spatial object and coordinate system

Use code to report the CRS, geometry type, bounding box, number of records, and first several coordinates in `occ_sf`. Compare the coordinates with those in `occ_projected`.

In code comments, briefly explain:

- what the CRS means;
- why the coordinate values change after `st_transform()` even though the locations do not move; and
- why a projected CRS is useful for the distance-based operation used to explore M.

Problem 2: Make and justify a map revision

Starting with `map_output`, make at least one change that improves the communication of the occurrence data. You may modify the mapped variable, point appearance, transparency, labels, map extent, or comparison shown. Save the revised map with a new descriptive filename in `output/`.

In your README, identify:

- what you changed;
- whether the change involved setting an aesthetic, mapping an aesthetic, changing a layer, or changing the viewing extent; and
- one way the choice could influence interpretation of the mapped occurrence data.

Problem 3: Explore area of accessibility

The 100-km buffer used in Part A represents one simple hypothesis about M. Create an **alternative hypothesis** by choosing a different buffer distance. Use `st_buffer()` and `st_union()` to create the alternative accessible area, plot it with the retained occurrence records, and save the figure in `output/`.

In your README, identify:

- What buffer distance did you use?
- How did changing the distance change the area represented as M?
- What biological assumption is implied by the distance you selected?
- Give **two reasons** a simple distance buffer might be a poor representation of the true M.
- What additional biological, geographic, or historical information would help you construct a more defensible M?

Problem 4: What about A and B?

We have created a spatial hypothesis about M, but we have not yet estimated A or B.

For *C. triseriatus*:

- identify one plausible abiotic factor that could contribute to **A**;
- identify one plausible biotic factor that could contribute to **B**; and
- for each, identify one type of data you would need before you could represent that component spatially.

Does either the occurrence map or the buffer map directly identify A or B? Explain.

Technical troubleshooting

Package installation fails. Copy the first error message into an Issue or note. Confirm that you are installing packages, not running `install.packages()` repeatedly inside your analysis script.

The map is blank or misplaced. Check `st_bbox(occ_sf)`, then inspect longitude and latitude ranges before changing the CRS.

A plot layer disappears. Check the order of your `geom_sf()` layers; later layers are drawn on top of earlier layers.

Help

Useful functions: `st_as_sf()`, `st_crs()`, `st_geometry()`, `st_geometry_type()`, `st_coordinates()`, `st_bbox()`, `st_transform()`, `st_buffer()`, `st_union()`, `ggplot()`, `geom_sf()`, `aes()`, `coord_sf()`, `labs()`, `theme_minimal()`, and `ggsave()`. Start with `?function_name`, run a small piece of code, and explain what the result means before moving on.

Academic integrity: The syllabus permits AI tools for learning and troubleshooting, but the submitted code, reasoning, and interpretation must be your own and understandable to you. Briefly acknowledge any AI use in a comment or README note.

References

- Pebesma, E. (2018). Simple Features for R: Standardized Support for Spatial Vector Data. *The R Journal*, 10(1), 439-446.
- Peterson, A. T., Soberon, J., Pearson, R. G., Anderson, R. P., Martinez-Meyer, E., Nakamura, M., & Araujo, M. B. (2011). *Ecological Niches and Geographic Distributions*. Princeton University Press. Chapter 3.

Assignment 2 - submit by Friday at midnight

Submit your private GitHub repository link within the Canvas assignment. The repository must contain the following items.

- **Reproducible spatial script.** `analysis/02_spatial_occurrence_map.R` runs from the project root and creates the map in `output/`.
- **Spatial object checks.** The script inspects the CRS, POINT geometry, bounding box, and coordinates before and after projection; your comments explain what the coordinate systems mean and why projection is useful for the distance-based analysis.
- **Mapped occurrence evidence.** At least one revised and clearly labeled `sf` occurrence map is saved in `output/`.
- **M hypothesis.** A second map shows an alternative distance-based hypothesis about M.
- **BAM reasoning.** The README explains why the buffer is only a hypothesis about M and identifies what additional evidence would be needed to represent A and B.
- **Version history.** Commits make your progress legible and the current repository is pushed to GitHub.

Add a new Lab 2 section to your README. Include: (1) one mapping decision and its consequence, (2) your alternative hypothesis for M and the buffer distance you chose, (3) why that buffer should not be interpreted as the true M, (4) what information would make M more defensible, and (5) what data would be needed to represent A and B.